Lec 02 Bellman Equation

Motivating Examples
- 计算Return

State Value
- **single-step process**
- multi-step trajectory
- Definition
- The relationship between return and state value

Bellman Equation: Derivation
- Bellman Equation
  - **Highlights:**

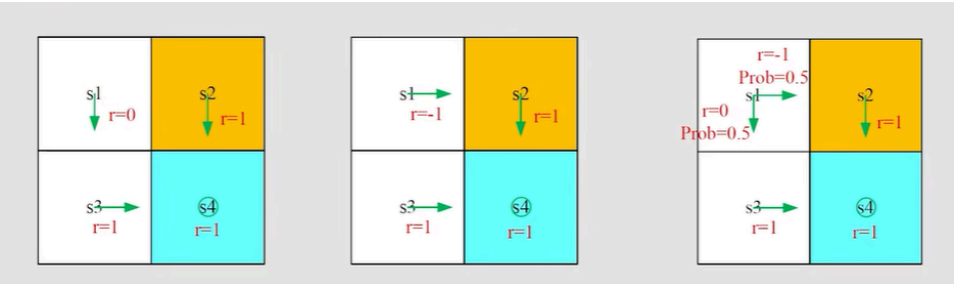
Bellman equation: Matrix-vector form
- Example

Bellman equation: Solve the state values
- The Closed-form solution
- An iterative solution
  - proof

Action Value
- Definition
- Example

Motivating Examples

return: The (discounted) sum of the rewards obtained along a trajectory.



Return could be used to evaluate policies.

- Based on policy 1:

$$
\begin{aligned}
return_1 &= 0 + \gamma 1 + \gamma^2 1 + \ldots \\
&= \gamma(1 + \gamma + \gamma^2 + \ldots) \\
&= \frac{\gamma}{1-\gamma}
\end{aligned}
$$

- Based on policy 2:

$$
\begin{aligned}
return_2 &= -1 + \gamma 1 + \gamma^2 1 + \ldots \\
&= -1 + \gamma(1 + \gamma + \gamma^2 + \ldots) \\
&= -1 + \frac{\gamma}{1-\gamma}
\end{aligned}
$$

- Based on policy 3:

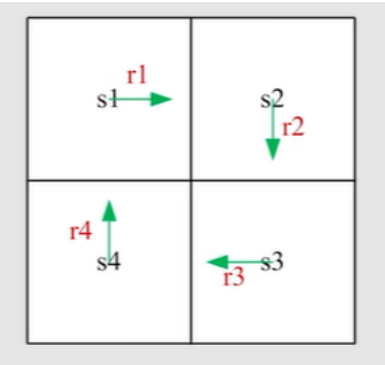
$$
\begin{aligned}
return_3 &= 0.5(-1 + \frac{\gamma}{1-\gamma}) + 0.5(\frac{\gamma}{1-\gamma}) \\
&= -0.5 + \frac{\gamma}{1-\gamma}
\end{aligned}
$$

In summary, starting from s_1,

$$
return_1 > return_3 > return_2
$$

> **Calculating return is important to evaluate a policy**

计算Return



Method 1:

$$
\begin{aligned}
v_1 &= r_1 + \gamma r_2 + \gamma^2 r_3 + \ldots \\
v_2 &= r_2 + \gamma r_3 + \gamma^2 r_4 + \ldots \\
v_3 &= r_3 + \gamma r_4 + \gamma^2 r_1 + \ldots \\
v_4 &= r_4 + \gamma r_1 + \gamma^2 r_2 + \ldots
\end{aligned}
$$

Method 2:

$$
\begin{aligned}
v_1 &= r_1 + \gamma(r_2 + \gamma r_3 + \dots) = r_1 + \gamma v_2 \\
v_2 &= r_2 + \gamma(r_3 + \gamma r_4 + \dots) = r_2 + \gamma v_3 \\
v_3 &= r_3 + \gamma(r_4 + \gamma r_1 + \dots) = r_3 + \gamma v_4 \\
v_4 &= r_4 + \gamma(r_1 + \gamma r_2 + \dots) = r_4 + \gamma v_1
\end{aligned}
$$

Write in the following matrix-vector form:

$$
\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}}_{\mathbf{v}} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} + \begin{bmatrix} \gamma v_2 \\ \gamma v_3 \\ \gamma v_4 \\ \gamma v_1 \end{bmatrix} = \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix}}_{\mathbf{r}} + \gamma \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{P}} \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}}_{\mathbf{v}}
$$

which can be rewritten as the **Bellman equation**

$$
\mathbf{v} = \mathbf{r} + \gamma \mathbf{P} \mathbf{v}
$$

State Value

single-step process

$$
S_t \xrightarrow{A_t} R_{t+1}, S_{t+1}
$$

- $t, t+1$: discrete time instances
- S_t : state at time t
- A_t : the action taken at state S_t
- R_{t+1} : the reward obtained after taking A_t
  （有时也记作R_t，但习惯上一般记作R_{t+1}）
- S_{t+1} : the state transited to after taking A_t

Note that S_t, A_t, R_{t+1} are all ***random variables***.（random variables 的意思是可以求一系列操作，比如 expectation（求期望））

This step is governed by the following probability distributions:

- $S_t \to A_t$ is governed by $\pi(A_t = a | S_t = s)$ **（是由策略决定的）**

- $S_t, A_t \to R_{t+1}$ is governed by $p(R_{t+1} = r | S_t = s, A_t = a)$ **（是由 reward probability 决定的）**
- $S_t, A_t \to S_{t+1}$ is governed by $p(S_{t+1} = s' | S_t = s, A_t = a)$ **（是由 state transition probability 决定的）**

At this moment, we assume we know the model (i.e., the probability distributions)!

multi-step trajectory

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, \ldots$$

The **discounted return** is

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots$$

- γ in $[0, 1)$ is a discount rate.
- G_t is also a random variable since $R_{t+1}, R_{t+2}, \ldots$ are ***random variables***.

Definition

The expectation (or called expected value or mean) of G_t is defined as the **state-value function** or simply **state value**:

$$v_\pi(s) = \mathbb{E}[G_t | S_t = s]$$

> $state\ value$ 就是在给定状态s下， 得到的 $return$均值。

Remarks:

- **It is a function of s**. It is a conditional expectation with the condition that the state starts from s.
- It is based on the policy π. **For a different policy, the state value may be different**.
- It represents the "value" of a state. If the state value is greater, then the policy is better because greater cumulative rewards can be obtained.
  state value 不仅仅是一个数值的 value，也可以代表一个**"价值"**，也就是当这个 state value 比较大的时候，**就代表这个状态是比较有价值的**，因为从这个状态出发，可以得到更多的 return。

The relationship between return and state value

return 是针对单个 trajectory 而言的，而 state value 则是多个 return 求平均值；如果从一个状态s 出发，可以得到多个 $return$，那么此时$return$和 state value 就是有区别的

Bellman Equation: Derivation

Bellman equation 描述了不同状态之间 state value 的关系

Consider a random trajectory:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, S_{t+3}, \ldots$$

The return can be written as:

$$\begin{aligned} G_t &= R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots \\ &= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \ldots) \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

Then, it follows from the definition of the state value that

$$\begin{aligned} v_\pi(s) &= \mathbb{E}[G_t | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s] \\ &= \mathbb{E}[R_{t+1} | S_t = s] + \gamma \mathbb{E}[G_{t+1} | S_t = s] \end{aligned}$$

First, calculate the first term $\mathbb{E}[R_{t+1} | S_t = s]$

$$\begin{aligned} \mathbb{E}[R_{t+1} | S_t = s] &= \sum_a \pi(a|s) \mathbb{E}[R_{t+1} | S_t = s, A_t = a] \\ &= \sum_a \pi(a|s) \sum_r p(r|s,a) r \end{aligned}$$

This is the mean of *immediate rewards.*

Second, calculate the second term $\mathbb{E}[G_{t+1} | S_t = s]$

$$
\begin{aligned}
\mathbb{E}[G_{t+1}|S_t=s] &= \sum_{s'} \mathbb{E}[G_{t+1}|S_t=s, S_{t+1}=s']p(s'|s) \\
&= \sum_{s'} \mathbb{E}[G_{t+1}|S_{t+1}=s']p(s'|s) \\
&= \sum_{s'} v_\pi(s')p(s'|s) \\
&= \sum_{s'} v_\pi(s') \sum_a p(s'|s,a)\pi(a|s)
\end{aligned}
$$

- **This is the mean of *future rewards*.**
- $\mathbb{E}[G_{t+1}|S_t=s, S_{t+1}=s'] = \mathbb{E}[G_{t+1}|S_{t+1}=s']$ due to **memoryless Markov property**

Bellman Equation

$$
\begin{aligned}
v_\pi(s) &= \mathbb{E}[R_{t+1}|S_t=s] + \gamma \mathbb{E}[G_{t+1}|S_t=s], \\
&= \underbrace{\sum_a \pi(a|s) \sum_r p(r|s,a)r}_{\text{mean of immediate rewards}} + \underbrace{\gamma \sum_a \pi(a|s) \sum_{s'} p(s'|s,a)v_\pi(s')}_{\text{mean of future rewards}}, \\
&= \sum_a \pi(a|s) \left[\sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a)v_\pi(s') \right], \quad \forall s \in \mathcal{S}.
\end{aligned}
$$

Highlights:

1. Bellman Equation 描述了不同 state value 之间的关系
2. 包括两部分：immediate reward term 和 future reward term
3. Bellman Equation 是一系列 equations，不是一个式子
4. 通过 bootstrapping 计算 $v_\pi(s)$ 和 $v_\pi(s')$
5. $\pi(a|s)$ 是给定的 policy，bellman equation依赖于policy，因此 解Bellman Equation 被称为 evaluate policy
6. $p(r|s,a)$ and $p(s'|s,a)$ represent the dynamic model，在这里假设我们知道这个model，后续有不知道的求解方法

Bellman equation: Matrix-vector form

对Bellman Equation

$$v_\pi(s)=\sum_a \pi(a|s)\left[\sum_r p(r|s,a)r+\gamma\sum_{s'}p(s'|s,a)v_\pi(s')\right]$$

变形为:

$$v_\pi(s)=r_\pi(s)+\gamma\sum_{s'}p_\pi(s'|s)v_\pi(s')$$

其中

$$\begin{aligned}r_\pi(s)&\triangleq\sum_a\pi(a|s)\sum_r p(r|s,a)r,\\ p_\pi(s'|s)&\triangleq\sum_a\pi(a|s)p(s'|s,a)\end{aligned}$$

将states记为 $s_i(i=1,\ldots,n)$

对state s_i 有

$$v_\pi(s_i)=r_\pi(s_i)+\gamma\sum_{s_j}p_\pi(s_j|s_i)v_\pi(s_j)$$

联立这些等式，重写为矩阵形式:

$$v_\pi=r_\pi+\gamma P_\pi v_\pi$$

其中

$v_\pi=[v_\pi(s_1),\ldots,v_\pi(s_n)]^T\in\mathbb{R}^n$

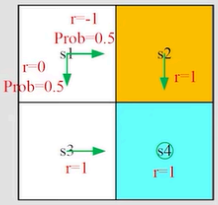
$r_\pi=[r_\pi(s_1),\ldots,r_\pi(s_n)]^T\in\mathbb{R}^n$

$P_\pi\in\mathbb{R}^{n\times n}$, where$[P_\pi]_{ij}=p_\pi(s_j|s_i)$, is the state transition matrix

Example

If there are four states, $v_\pi = r_\pi + \gamma P_\pi v_\pi$ can be written out as

$$\underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi} = \underbrace{\begin{bmatrix} r_\pi(s_1) \\ r_\pi(s_2) \\ r_\pi(s_3) \\ r_\pi(s_4) \end{bmatrix}}_{r_\pi} + \gamma \underbrace{\begin{bmatrix} p_\pi(s_1|s_1) & p_\pi(s_2|s_1) & p_\pi(s_3|s_1) & p_\pi(s_4|s_1) \\ p_\pi(s_1|s_2) & p_\pi(s_2|s_2) & p_\pi(s_3|s_2) & p_\pi(s_4|s_2) \\ p_\pi(s_1|s_3) & p_\pi(s_2|s_3) & p_\pi(s_3|s_3) & p_\pi(s_4|s_3) \\ p_\pi(s_1|s_4) & p_\pi(s_2|s_4) & p_\pi(s_3|s_4) & p_\pi(s_4|s_4) \end{bmatrix}}_{P_\pi} \underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi}.$$



For this specific example:

$$\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix} = \begin{bmatrix} 0.5(0) + 0.5(-1) \\ 1 \\ 1 \\ 1 \end{bmatrix} + \gamma \begin{bmatrix} 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}.$$

Bellman equation: Solve the state values

- Why to solve Bellman Equation?
  对于给定的policy，只有找到其对应的state value，才能进行policy evaluation，即对policy进行评估，后续才能根据结果进行改进。

The Closed-form solution

$$v_\pi = (I - \gamma P_\pi)^{-1} r_\pi$$

实际我们并不会采用这个方法，当状态空间比较大的时候，这个矩阵的维数很大，求逆的计算量很大。

An iterative solution

$$v_{k+1} = r_\pi + \gamma P_\pi v_k$$

将v_0代入v_k得到v_1，将v_1代入v_k得到v_2，以此类推，可以证明：

$$v_k \to v_\pi = (I - \gamma P_\pi)^{-1} r_\pi, \quad k \to \infty$$

proof

Proof.

Define the error as $\delta_k = v_k - v_\pi$. We only need to show $\delta_k \to 0$. Substituting $v_{k+1} = \delta_{k+1} + v_\pi$ and $v_k = \delta_k + v_\pi$ into $v_{k+1} = r_\pi + \gamma P_\pi v_k$ gives

$$\delta_{k+1} + v_\pi = r_\pi + \gamma P_\pi(\delta_k + v_\pi),$$

which can be rewritten as

$$\delta_{k+1} = -v_\pi + r_\pi + \gamma P_\pi \delta_k + \gamma P_\pi v_\pi = \gamma P_\pi \delta_k.$$

As a result,

$$\delta_{k+1} = \gamma P_\pi \delta_k = \gamma^2 P_\pi^2 \delta_{k-1} = \cdots = \gamma^{k+1} P_\pi^{k+1} \delta_0.$$

Note that $0 \le P_\pi^k \le 1$, which means every entry of P_π^k is no greater than 1 for any $k = 0, 1, 2, \ldots$. That is because $P_\pi^k \mathbf{1} = \mathbf{1}$, where $\mathbf{1} = [1, \ldots, 1]^T$. On the other hand, since $\gamma < 1$, we know $\gamma^k \to 0$ and hence $\delta_{k+1} = \gamma^{k+1} P_\pi^{k+1} \delta_0 \to 0$ as $k \to \infty$. □

Action Value

- **State value: the average return the agent can get starting from a state**
- **Action value: the average return the agent can get starting from a state and taking an action**

Definition

$$q_\pi(s, a) = \mathbb{E}[G_t | S_t = s, A_t = a]$$

- $q_\pi(s, a)$ 是一个 state-action pair (s, a) 的函数
- $q_\pi(s, a)$ 依赖于策略 π

$$\underbrace{\mathbb{E}[G_t | S_t = s]}_{v_\pi(s)} = \sum_a \underbrace{\mathbb{E}[G_t | S_t = s, A_t = a]}_{q_\pi(s,a)} \pi(a|s) \quad (1)$$

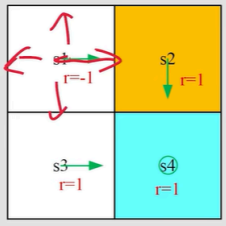
Hence

$$v_\pi(s) = \sum_a \pi(a|s) q_\pi(s, a)$$

$$q_\pi(s,a) = \sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a)v_\pi(s') \quad (2)$$

- **(1) shows how to obtain state values from action values**
- **(2) shows how to obtain action values from state values**

Example



Write out the action values for state s_1.

$$q_\pi(s_1, a_2) = -1 + \gamma v_\pi(s_2),$$

Questions:

- $q_\pi(s_1, a_1), q_\pi(s_1, a_3), q_\pi(s_1, a_4), q_\pi(s_1, a_5) =?$ Be careful!

- $q_\pi(s_1, a_1), q_\pi(s_1, a_3), q_\pi(s_1, a_4), q_\pi(s_1, a_5) \neq 0$

For the other actions:

$$\begin{aligned}
q_\pi(s_1, a_1) &= -1 + \gamma v_\pi(s_1), \\
q_\pi(s_1, a_3) &= 0 + \gamma v_\pi(s_3), \\
q_\pi(s_1, a_4) &= -1 + \gamma v_\pi(s_1), \\
q_\pi(s_1, a_5) &= 0 + \gamma v_\pi(s_1).
\end{aligned}$$

- 可以先将所有的 state value 都计算出来，再计算 action value；
- 也可以不用计算 state value ，直接计算 action value ，比如通过数据的方式，此时就不再依赖于这个模型了